

# Entropic metastability in the narrow escape problem

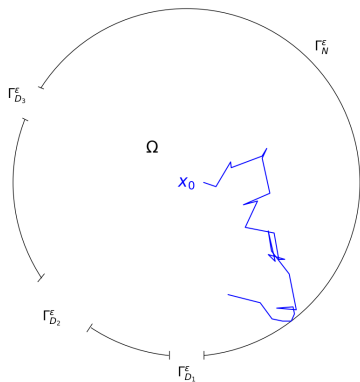
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# The narrow escape problem

**Rare event:** metastability induced by **entropic** barriers ( $\neq$  energetic)



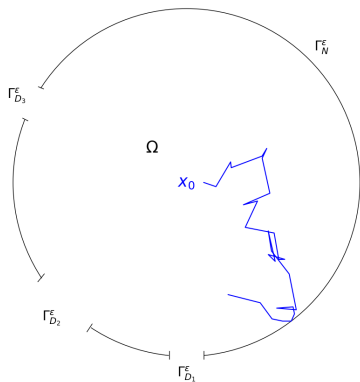
## Setting:

- Domain  $\Omega$  with holes  $\Gamma_D^\varepsilon$  and **reflecting boundary**  $\Gamma_N^\varepsilon$
- A **Brownian motion** starting at  $x_0$  taking a **long time** to exit  $\tau_\varepsilon = \inf\{t \geq 0 \mid X_t \notin \overline{\Omega}\}$

$$dX_t = \sqrt{2} dB_t - \mathbb{1}_{\Gamma_D^\varepsilon}(X_t) n(X_t) dL_t$$

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**Goal:** In the limit of **small holes**  $\varepsilon \rightarrow 0$ :

- The law of the escape time  $\tau_\varepsilon$
- The law of exit hole  $X_{\tau_\varepsilon}$

**Motivation:** Entropic transitions, **Biology:** Ion channels, diffusion in cells

# From the narrow escape problem to an eigenvalue problem

## Key object:

the quasi-stationary distribution

$$\nu_\varepsilon := \lim_{t \rightarrow +\infty} \text{Law}(X_t \mid t < \tau_\varepsilon)$$

In the limit  $\varepsilon \rightarrow 0$ ,

$X_t$  reaches  $\nu_\varepsilon$  long before it escapes

If  $X_0 \sim \nu_\varepsilon$ :

- $\tau_\varepsilon \sim \text{Exp}(\lambda_\varepsilon)$  independent of  $X_{\tau_\varepsilon}$
- $X_{\tau_\varepsilon} \sim \partial_n \nu_\varepsilon|_{\Gamma_D^\varepsilon}$

Learn more in: Di Gesù, Lelièvre, Le Peutrec and Nectoux (2016)

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The QSD  $\nu_\varepsilon$  is the solution of

$$\begin{cases} -\Delta \nu_\varepsilon = \lambda_\varepsilon \nu_\varepsilon & \text{in } \Omega \\ \partial_n \nu_\varepsilon = 0 & \text{on } \Gamma_N^\varepsilon \\ \nu_\varepsilon = 0 & \text{on } \Gamma_D^\varepsilon \end{cases}$$

with  $\lambda_\varepsilon > 0$  the smallest eigenvalue of the above problem

- Use of a quasi-mode ansatz to get the asymptotic expansion of  $\nu_\varepsilon$  in the limit  $\varepsilon \rightarrow 0$
- In any dimension  $d \geq 2$ , and smooth domains with  $N$  holes

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## Results on the exit time

### Theorem [Asymptotic of the exit time]

Consider only **one hole** of radius  $r_\varepsilon$ . Then there exists a  $C_{d,\Omega} > 0$  such that the eigenvalue  $\lambda_\varepsilon$  scales as:

$$\lambda_\varepsilon = \left(\mathbb{E}[\tau_\varepsilon]\right)^{-1} = \begin{cases} C_{d,\Omega} r_\varepsilon^{d-2} & + O(r_\varepsilon^{d-1}), & \text{for } d > 3 \\ C_{3,\Omega} r_\varepsilon & + O(r_\varepsilon^2 \log(r_\varepsilon)), & \text{for } d = 3 \\ C_{2,\Omega} (\log(r_\varepsilon))^{-1} & + O([\log(r_\varepsilon)]^{-2}), & \text{for } d = 2 \end{cases}$$

Where does the scaling come from? (Potential theory [1, 2])

The fundamental solution of the laplacian  $\Delta$  in dimension  $d$ :

$$\lambda_\varepsilon \sim C_{d,\Omega} \Lambda(r_\varepsilon)^{-1} \quad \text{and} \quad C_{d,\Omega} = \frac{\max\{d-2, 1\}}{2} \frac{|\partial B(0, 1)|}{|\Omega|}$$

[1] Ammari, Kalimeris, Kang and Lee (2010), [2] Felli, Noris and Ognibene(2020)